

M4 Exploratory Data Analytics Report

Traffic Accidents Dataset

Team Name

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1 Introduction

This report presents an exploratory data analysis (EDA) of a cleaned traffic accidents dataset. The goal of this milestone is to understand the structure of the data, identify important patterns, examine relationships between variables, revise initial hypotheses, and define a clear modelling question for the next stage of the project.

Each row in the dataset represents one traffic accident record. The dataset includes accident context, environmental conditions, road characteristics, crash type, number of units involved, and injury severity. The EDA is organised into univariate, bivariate, and multivariate analyses. Each visualisation is interpreted with a specific takeaway so that the figures support later modelling decisions rather than only describing the data visually.

2 Dataset Overview and Data Quality

The cleaned dataset contains 209,275 records and 17 columns. There are no missing values in the cleaned version. Most variables are categorical, including `traffic_control_device`, `weather_condition`, `lighting_condition`, `first_crash_type`, `trafficway_type`, `roadway_surface_cond`, `crash_type`, and `most_severe_injury`. The main numeric variable is `num_units`, which records the number of units involved in each accident.

Table 1: Dataset overview

Item	Value
Number of records	209,275
Number of columns	17
Missing values	0
Main numeric variable	<code>num_units</code>
Main outcome candidates	<code>crash_type</code> , <code>most_severe_injury</code>
Main feature groups	Weather, lighting, roadway, traffic control, crash context

The cleaned dataset also includes engineered quality indicators such as `flag_num_units_high_outlier`. These flags are useful for data checking, but they should be handled carefully during modelling because some may duplicate information already present in the original variables.

3 Univariate Analysis

Univariate analysis examines individual variables before comparing them with other features. This section focuses on the distribution of `num_units` and the class balance of `crash_type`.

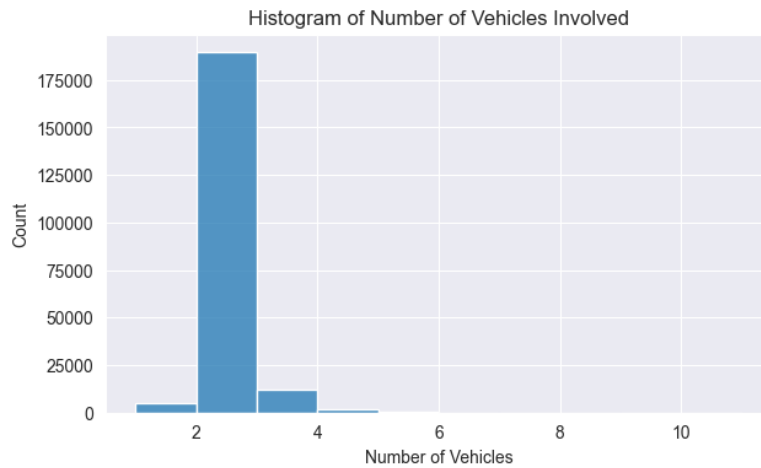


Figure 1: Distribution of number of units involved in crashes

Interpretation. Figure 1 shows that most accidents involve two units. Crashes involving more than three units are rare, and very high values occur only in a small number of records. This indicates that `num_units` is highly concentrated around typical two-unit crashes rather than evenly distributed across many possible values.

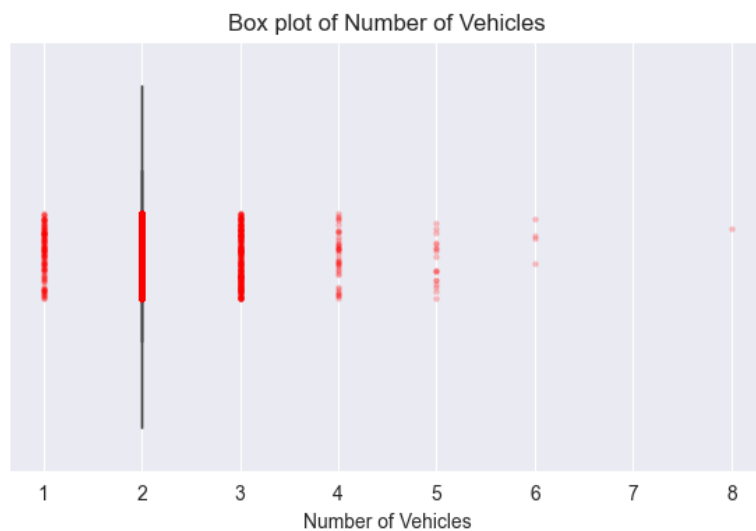


Figure 2: Box plot of number of units involved

Interpretation. Figure 2 confirms that the typical accident involves two units. Values above the usual range represent unusual multi-unit crashes. This supports the use of outlier checks for `num_units`, but it also shows that most observations are concentrated in a very narrow range, which may limit how much this variable can explain by itself.

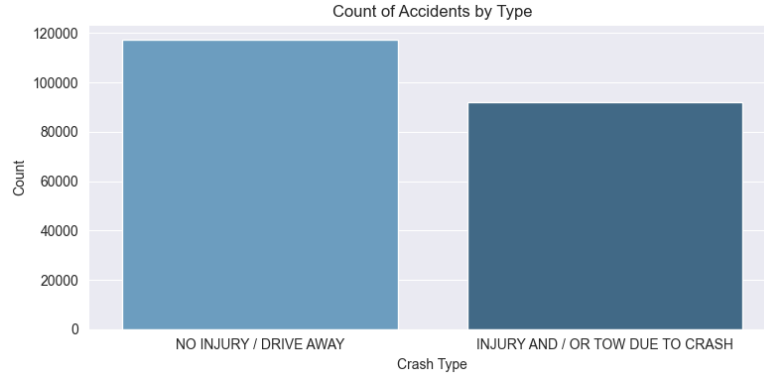


Figure 3: Crash count by crash type

Interpretation. Figure 3 shows that NO INJURY / DRIVE AWAY crashes are more common than INJURY AND / OR TOW DUE TO CRASH. This indicates a class imbalance in the potential target variable. If `crash_type` is used as the modelling outcome, accuracy alone may be misleading, because a model could perform well by mainly predicting the majority class.

4 Bivariate Analysis

Bivariate analysis examines relationships between two variables. This section focuses on how the number of units involved differs across crash types.

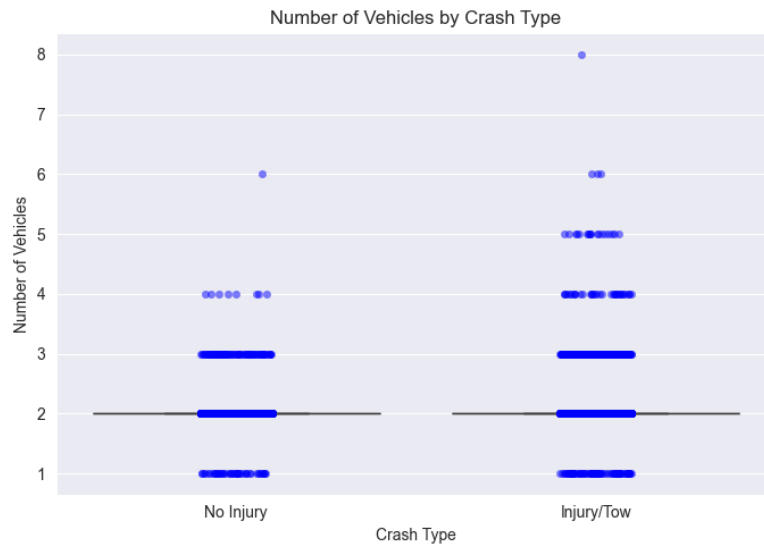


Figure 4: Number of units by crash type

Interpretation. Figure 4 compares the distribution of `num_units` between the two crash-type categories. Both categories are centred around two units, which means that a typical accident involves two units regardless of crash type. However, the injury/tow category has a slightly longer upper tail, suggesting that more serious crash types are somewhat more likely to involve additional units.

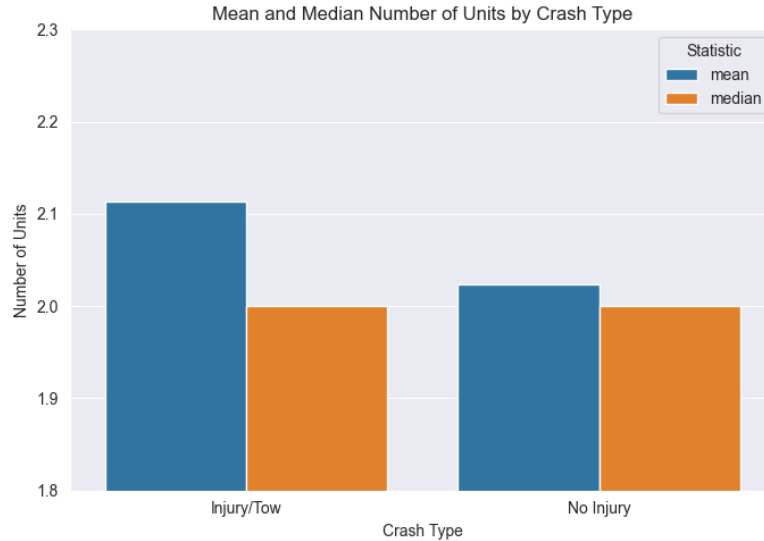


Figure 5: Mean and median number of units by crash type

Interpretation. Figure 5 supports the previous finding numerically. The median number of units is about two for both crash types, while the mean is slightly higher for INJURY AND / OR TOW DUE TO CRASH. This suggests that additional units are more common in the more serious crash category, but the difference is modest. Therefore, `num_units` may be useful, but it is unlikely to be sufficient as a single predictor.

5 Multivariate Analysis

Multivariate analysis considers patterns involving multiple variables. This is important because crash outcomes are unlikely to be explained by only one factor. Instead, accident severity and crash type may depend on vehicle involvement, environmental conditions, road context, and traffic control features.

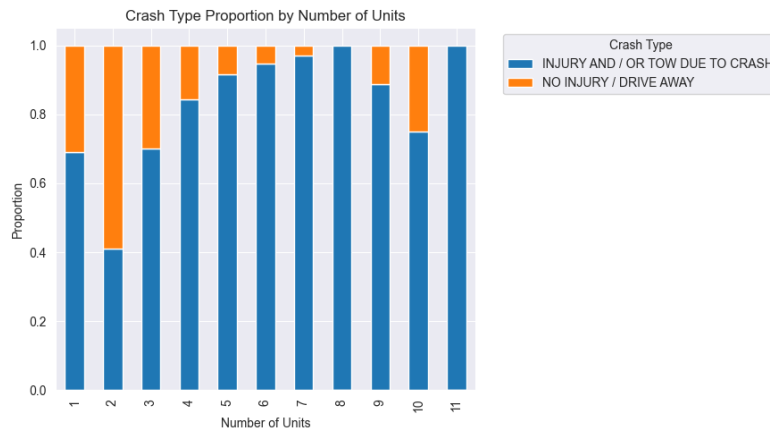


Figure 6: Proportion of number of units by injury severity

Interpretation. Figure 6 shows that two-unit crashes dominate every injury severity category. However, the proportions are not identical across severity levels. Fatal crashes have relatively higher shares of one-unit and three-unit crashes compared with no-injury crashes. This suggests that `num_units` is related to injury severity, but the relationship is not simply “more units means more severe”. Injury severity likely depends on multiple accident and environmental factors.

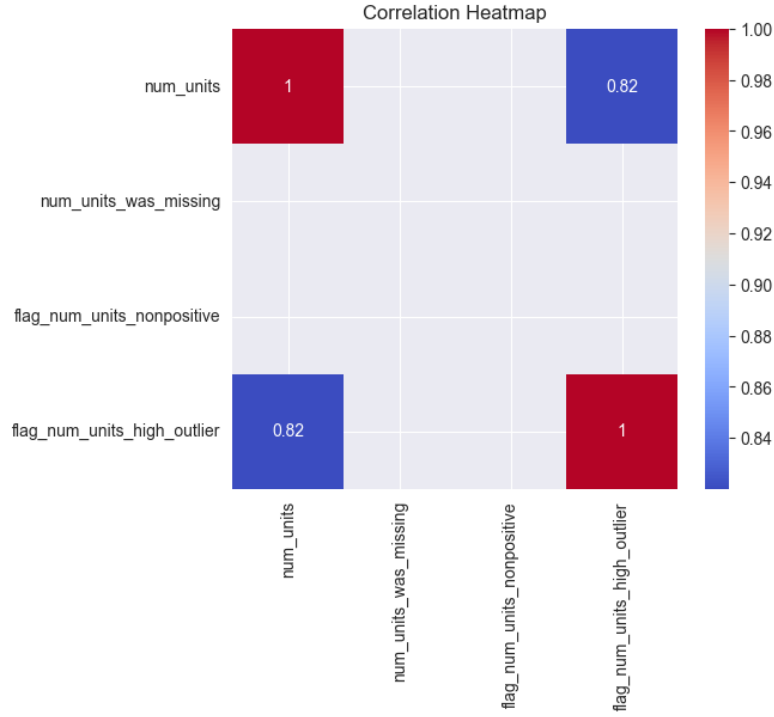


Figure 7: Crash type proportion by number of units

Interpretation. Figure 7 shows how crash-type proportions change as the number of units increases. Crashes with higher unit counts tend to have a larger share of INJURY AND / OR TOW DUE TO CRASH. This supports the idea that multi-unit accidents may be more likely to involve injury or towing, although the number of units alone does not fully determine crash outcome.

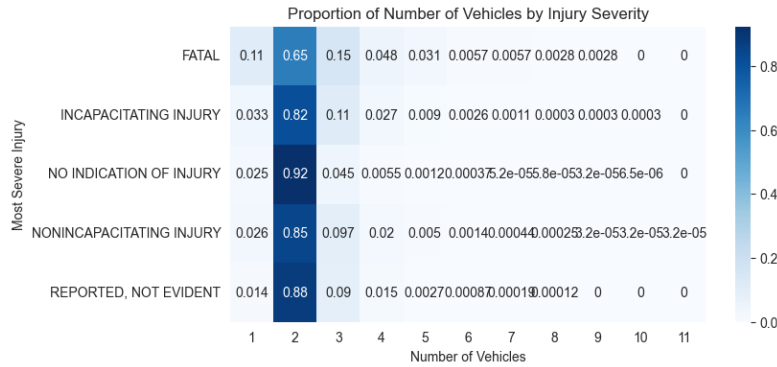


Figure 8: Correlation heatmap for numeric variables and flags

Interpretation. Figure 8 shows a strong positive correlation between `num_units` and `flag_num_units_high_outlier`. This is expected because the flag is derived from unusually high values of `num_units`. The key modelling implication is that engineered flags should be checked for redundancy. Including both the original variable and a directly derived flag may not add much new information and could affect model interpretation.

6 Revised Hypotheses

The EDA findings refined the initial hypotheses as follows:

Table 2: Revised hypotheses based on EDA

Initial Hypothesis	EDA Finding	Revised Hypothesis
Crashes with more units are more severe.	Two-unit crashes dominate all injury severity levels, but fatal crashes show a slightly different unit distribution.	<code>num_units</code> may contribute to injury severity, but it should be combined with crash type and environmental features.
Injury/tow crashes involve more units than no-injury crashes.	The median is about two units for both crash types, but the injury/tow group has a slightly higher mean and longer tail.	More serious crash types are somewhat more likely to involve additional units, but the effect is modest.
Engineered flags directly improve modelling.	The high-outlier flag is strongly correlated with <code>num_units</code> .	Engineered flags should be tested for redundancy before being included as model features.
A single variable can explain crash outcome.	No single EDA plot shows a complete separation of crash outcomes.	Crash outcome should be modelled using multiple features, including number of units, weather, lighting, roadway condition, intersection status, and contributory cause.

7 Modelling Question

Based on the EDA, the team will pursue the following modelling question:

Can we predict whether a traffic accident results in **INJURY AND / OR TOW DUE TO CRASH** using features such as number of units involved, traffic control device, weather condition, lighting condition, roadway surface condition, intersection status, and primary contributory cause?

This modelling question is appropriate because `crash_type` is a binary outcome, making it suitable for classification. The EDA suggests that `num_units` is relevant but insufficient by itself. Therefore, the model should combine accident context, road conditions, and environmental variables. Because the target distribution is imbalanced, model evaluation should include precision, recall, F1-score, and confusion matrices rather than relying only on accuracy.

8 Conclusion

The EDA shows that most traffic accidents in the cleaned dataset involve two units and fall into the no-injury/drive-away category. Injury/tow crashes have a slightly higher average number of units and a longer upper tail, but the relationship between vehicle count and severity is not simple. Two-unit crashes dominate all injury severity levels, while fatal crashes show somewhat different unit distributions. These findings suggest that crash outcomes are influenced by multiple factors rather than a single variable. The next stage should therefore focus on building a classification model for `crash_type`, while carefully handling class imbalance and avoiding redundant features.